BIRMINGHAM CITY
University

CMP6200/DIG6200
Individual Undergraduate Project (FYP)
Interim Progress Review
AR based Virtual Tour Guide

Student Name: Sadikshya Ghimire

Student ID: 23140745

Course: CMP6200 Individual Honours Project

Supervisor: Sujan Shrestha

Reporting Period: 19 December, 2025 to January 29,2026

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1. Log of Completed Activities & Milestones

This report is the compilation of all the technical and non technical work that I have done during the period of the review of 19 December 2025 to 29 January 2026.

Major tasks completed were:

Revising the topic: I chose the project topic and scope as a Smart Virtual Tour Guide with the help of Computer Vision, Augmented Reality and NLP.
Definition & Objectives: I have stated three problem statements and declared SMART objectives that fit the technical and academic requirements.
Literature Review: Updated available literature on AR in cultural tourism, landmark recognition by computer vision and tourism recommendation systems.
Methodology and Scope Definition: I have stated how in my project Unity, a based AR, CV, based landmark recognition and API will collaborate in NLP chatbot integration.
Interim Report Preparation & Submission: I Filled the interim project report according to the official template and guidelines.
Project Planning: I Have drawn an elaborate Gantt chart with all the phases, jobs, and milestones during the project schedule.
Preliminary Video Data Collection: I have filmed the short videos of the heritage landmarks in Kathmandu to the purpose of data set.
Image Frame Extraction: Frames were extracted successfully in videos that were recorded.
Phase Two: Began performing image preprocessing and data augmentation, planning to train models.

2. Reflection and Iteration

This section shows the progress made so far and highlights the challenges and how they were resolved during the project lifecycle.

2.1 Progress Against Original Timeline

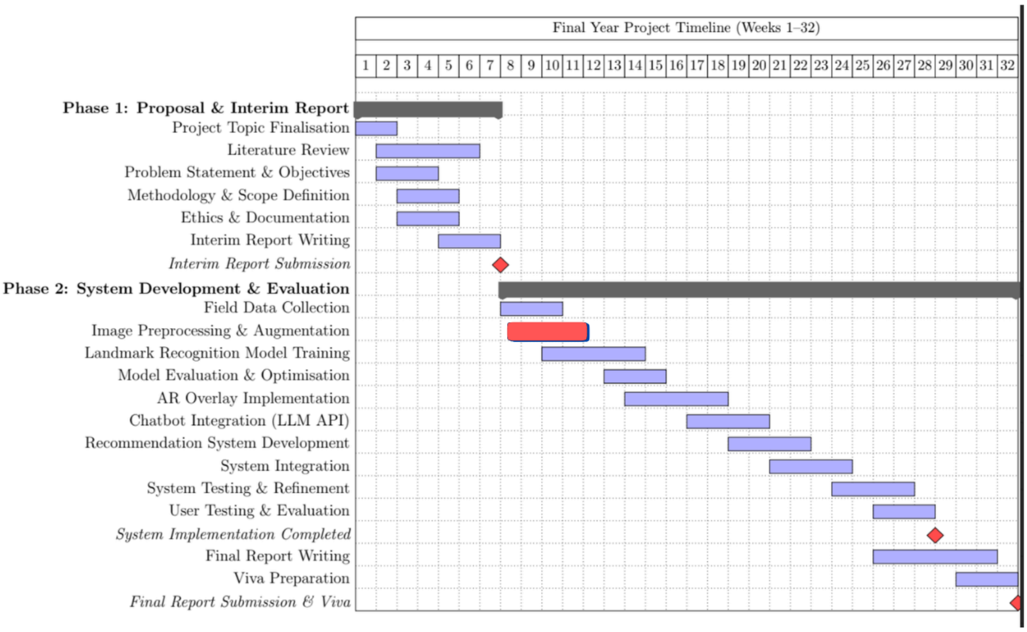

Figure 1: Updated Gantt Chart

Based on the updated Gantt chart and completed milestones, the project is currently:

☑ On Track

Image preprocessing and augmentation phase have been completed as per the original timeline. There are no delays that affect future milestones.

2.2 Challenges Encountered & Resolutions

While doing this, I faced many challenges, mainly related to technical issues and resource constraints. These challenges and resolved actions are summarised below:

Challenge Encountered	**Impact on Project**	**Action Taken / Planned Solution**
Initial lack of clarity regarding the scope and type of AR implementation	Temporary uncertainty in system design decisions.	Clarified AR scope by selecting contextual markerless AR info cards instead of 3D reconstruction, ensuring feasibility
Learning curve associated with Unity AR Foundation	Slower start to AR development	Allocated dedicated self-learning time using official Unity documentation and tutorials
Limited prior experience with AR development	Increased time required to understan d AR workflows	Broke AR development into smaller, manageable milestones and focused on basic AR functionality first
Dataset preparation taking longer than expected	Delayed start of model training	Adopted video-to-frame extraction approach to efficiently generate sufficient training images
Hardware and memory constraints on local machine	Occasional interruptions during preprocessing tasks	Optimised image resolution and planned to use institutional resources where required

Figure 2: Tabular description of encountered challenges and resolutions

3. Action Plan: Goals for Next Period

Works I will completed by next supervisor meeting (Feb 12, 2026):

- I will Complete image preprocessing and data augmentation for chosen landmarks.
- I will Test image quality and class balance.
- I will Start initial training of a lightweight model for landmark recognition that can be used in mobile devices.
- I will establish a basic Unity AR scene and test camera feed functionality using AR Foundation.

4. Other Considerations

4.1 Scope Adjustments and Clarifications

The field of the project following the first research and analysis has been shifted to a smaller one by choosing the ten heritage landmarks, deleting the aspect of 3D reconstruction due to the unavailability of time and computer power and Kathmandu Valley is the only region taken into account. The correction enables the likelihood of the project without it being lost in its academic side

5. Questions for Supervisor

1. Is the use of contextual AR information cards sufficient to demonstrate effective AR integration, or should limited interactive elements be included?
2. Which evaluation metrics would you recommend for assessing AR usability and user experience?
3. Are there any concerns regarding the current dataset size or the selection of heritage landmarks?

6. Conclusion and Reflection

At this time, the project was shifting into the initial phases of technical work. We reduced the number of landmarks and removed 3D reconstruction, instead we concentrated on contextual markerless AR. This helped reduce the size of the project and made it manageable, but still of academic value.Organized self-learning, better workflow, and realistic technical planning helped to fix the issues of the AR learning curve, the time wasted during dataset preparation, and hardware constraints. Altogether, this step has me to clarify the system architecture, reinforce the project direction and precondition the effective model training, development of AR and integration of all the systems further